Dilip Bhakadiwal

+91-8003046831 | 9828dilip@gmail.com | linkedin.com/in/dilip-bhakadiwal | github.com/Dilip-Bhakadiwal/AI-Projects | Jaipur, Rajasthan (Open to Remote)

AI/ML Engineer specializing in Agentic workflows, GraphRAG, and high-performance LLM orchestration. Proven experience architecting scalable multi-agent systems, deploying open-source models locally (Llama 3.3, Qwen), and optimizing retrieval pipelines for enterprise-grade, sub-second inference.

TECHNICAL SKILLS

Agentic AI & Orchestration: LangGraph, LangChain, Model Context Protocol (MCP), ReAct Agent Workflows
Backend & Cloud Pipelines: FastAPI, AWS, Docker, Render, GitHub Actions (CI/CD)
Data Processing & Retrieval: GraphRAG, Neo4j, Pinecone, PostgreSQL, FastEmbed, Redis
LLM Integration & Evals: LangSmith, Prompt Guardrails, PII Redaction
Programming & Tools: Python, PyTorch, TypeScript, React, Git, Pandas, NumPy

EDUCATION

Defence Institute of Advanced Technology (DIAT) — 2024 – 2026
Master of Technology in Artificial Intelligence – CGPA: 7.33 — Pune, Maharashtra

MBM University — 2019 – 2023
Bachelor of Engineering in Electronics and Computer Engineering | **GATE 2023 Qualified** — Jodhpur, Rajasthan

KEY AI ENGINEERING PROJECTS

Nexora AI: Enterprise GraphRAG Platform | *LangGraph, Neo4j, Pinecone, Groq LPU* — 2026

- Architected a multi-agent GraphRAG platform unifying Neo4j Cypher traversals with Pinecone semantic search, achieving 99.4% multi-hop retrieval accuracy.
- Engineered a sub-second (<200ms) LPU inference pipeline using Groq with automated failover, deployed as a fully containerized React and FastAPI architecture on Render.
- Implemented an ephemeral memory engine with active PII redaction guardrails (stripping SSNs, API keys) and strict rate-limiting for enterprise-grade data security.

AlignAI Engine: Autonomous Job Discovery | *LangGraph, Pinecone, Neo4j, FastEmbed* — 2026

- Developed a LangGraph-orchestrated multi-agent state machine to autonomously ingest and normalize API-sourced job postings, processing 100+ roles/minute and eliminating 85% of keyword-matching noise.
- Designed a 3-signal fit-ranking algorithm combining dense cosine similarity, Neo4j graph coverage, and LLM evaluation for explainable candidate-role alignment.
- Built a SHA-256 caching architecture utilizing RapidFuzz for continuous entity resolution, eliminating graph node duplication and automatically recomputing vectors on profile updates.

RESEARCH EXPERIENCE

Edge AI Object Detection System on FPGA and Jetson Platforms — Sept. 2025 – Jan. 2026
Research Project, DIAT — Pune, Maharashtra

- Designed and quantized (FP32 to INT8) a lightweight YOLOv8n architecture, achieving 45 FPS on an NVIDIA Jetson Orin and 13 FPS on a Xilinx FPGA for edge deployment.
- Integrated a LLaMA 1B model to generate contextual natural-language descriptions of detected objects in real time.

Focal-CBAM Fish-YOLO | *Funded by MoES, Govt. of India* — Aug. 2025 – Oct. 2025
Research Project, DIAT — Pune, Maharashtra

- Developed an attention-enhanced YOLOv8 detection architecture using a novel Focal-CBAM module to improve feature attention and accuracy in noisy underwater environments.
- Optimized multi-scale detection heads on the RUOD dataset, successfully outperforming baseline architectures on underwater localization benchmarks.

PUBLICATIONS

Deep Underwater Fish Detection via Focal Modulated Channel Attention in YOLO | *1st Author, ICASA 2026 (London)*

ANIMA: YOLOv8-Based Framework for Object Detection & Compression | *2nd Author, IEEE Xplore* (Link)